Case Study

Wine Quality Assessment Using Physicochemical Properties

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1. Problem Statement

The company currently judges winequality only through sensory assessment by tasters, which is

slow and subjective. Management wants to know whether laboratory (physicochemical) measurements can help explain differences in quality and support a data-driven screening process. The objectives are to (i) describe the main variables statistically, (ii) find which measurements are most strongly associated with quality, (iii) build and evaluate a multiple linear regression model that predicts the quality score, and (iv) advise whether the model can be used in practice.

Dataset. Red Vinho Verde wine from Portugal (Cortez et al., 2009): 1599 samples, 11 physicochemical inputs (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol) and one output, quality, which is the median of at least three expert scores. There are no missing values. Only the red wine file is analysed in this report.

2. Methodology

The work wasdoneinPython (pandas, NumPy, SciPy, scikit-learn, statsmodels, matplotlib, seaborn).
The steps were:

1) Load and check the data (shape, missing values, duplicates; 240 duplicate rows were found and kept because different bottles can have identical readings and scores).
2) Descriptive statistics (mean, median, mode, standard deviation, skewness) for alcohol, volatile acidity and residual sugar; the coefficient of variation (CV = std / mean x 100) to compare variability across different scales; Pearson's r for alcohol vs quality.
3) Exploratory analysis: histogram of quality, box plot of alcohol by quality, correlation heatmap.
4) Multiple linear regression on all 11 inputs, 80/20 train-test split with random_state = 42; evaluation with MAE, MSE, RMSE and R^2 on the test set.
5) Standardized coefficients to compare predictor influence, and Variance Inflation Factors (VIF) to check multicollinearity.
6) A predicted vs actual plot and a business interpretation.

3. Statistical Findings

3.1 Descriptive statistics (Task 1a, 1b)

Variable	Mean	Median	Mode	Std	Skewness	CV (%)
alcohol	10.423	10.200	9.500	1.066	0.861	10.22
volatile acidity	0.528	0.520	0.600	0.179	0.672	33.92
residual sugar	2.539	2.200	2.000	1.410	4.541	55.53

Alcohol has a mean (10.42) slightly above its median (10.2) and a skewness of 0.86, so it is moderately right skewed. **Volatile acidity** has almost equal mean and median (0.528 vs 0.52) and skewness 0.67, so it is fairly symmetric with a small right tail. **Residual sugar** has a skewness of 4.54: a few wines with a lot of sugar (maximum 15.5) pull the mean (2.54) well above the median (2.2). For all three variables mean > median, which is the typical sign of positive skew.

Standard deviations cannot be compared directly because the variables are on different scales, so the coefficient of variation is used. **Residual sugar has the greatest relative variability (CV about 55.5%)**, followed by volatile acidity (33.9%) and alcohol (10.2%). Alcohol has the largest standard deviation (1.07) only because it is measured on a bigger scale; relative to its mean it is the most stable.

3.2 Alcohol and quality (Task 1c)

Pearson's correlation between alcohol and quality is **r = +0.476** ($p < 0.001$). The direction is positive: wines with more alcohol tend to receive higher scores. The strength is moderate, and $r^2 = 0.23$ means alcohol alone explains about 23% of the variation in quality. With 1599 samples even small correlations are significant, so the size of r matters more than the p-value. This is an association, not proof that raising alcohol improves a wine.

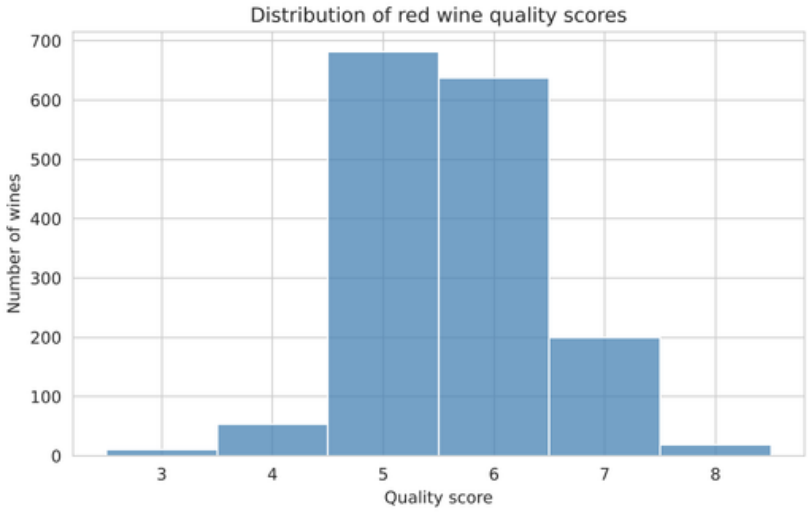


Figure 1: Histogram of quality scores (x-axis: quality score, y-axis: number of wines).

Interpretation of Figure 1. Scores only range from 3 to 8. Scores 5 (681 wines) and 6 (638 wines) make up about 83% of the data, while 3 and 8 are very rare (10 and 18 wines). The shape is roughly bell shaped and almost symmetric (skewness 0.22) but strongly peaked. The classes are imbalanced, so the model has very few examples of the best and worst wines to learn from.

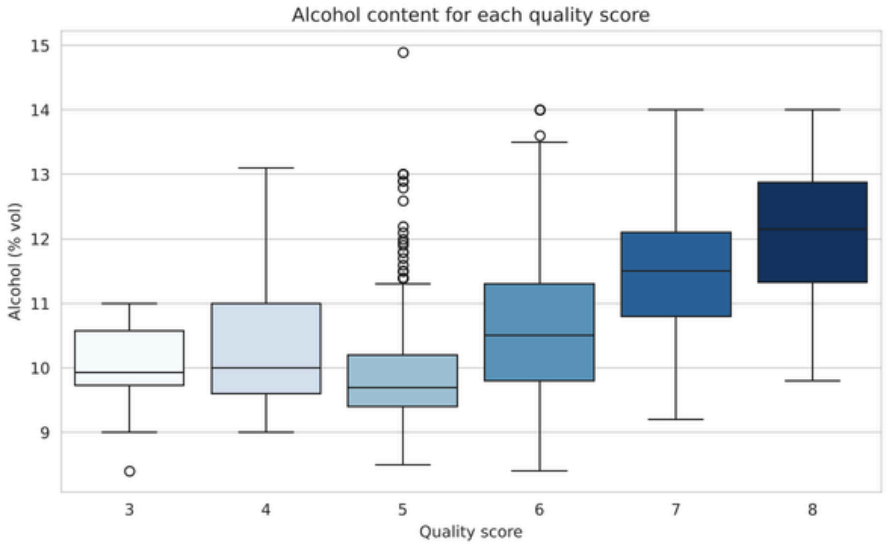


Figure 2: Box plot of alcohol (% vol) for each quality score.

Interpretation of Figure 2. From quality 5 upwards the median alcohol rises steadily: about 9.7 (quality 5), 10.5 (6), 11.5 (7) and 12.2 (8), which agrees with the positive correlation. At the low end the pattern is not steady (qualities 3 and 4 have medians of about 9.9 and 10.0, above quality 5), but those groups have only 10 and 53 wines. Using the 1.5 x IQR rule, quality 5 has the most outliers (30, all high alcohol values between 11.4 and 14.9), quality 6 has 5 high outliers (13.6 to 14.0) and quality 3 has one low value (8.4). So many high-alcohol wines still received only an average score, and the boxes of neighbouring scores overlap a lot, so alcohol alone cannot separate them cleanly.

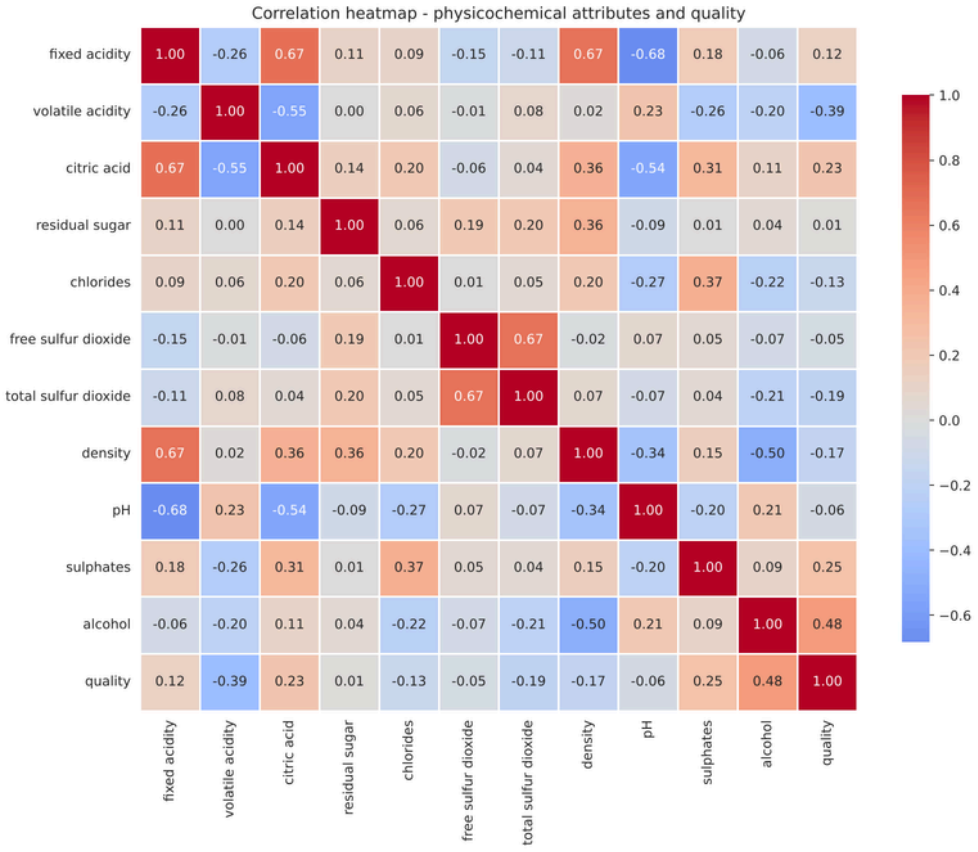


Figure 3: Correlation heatmap of the 11 physicochemical attributes and quality.

Interpretation of Figure 3. Besides the quality row, several inputs are strongly related to each other: fixed acidity with citric acid (+0.67), density (+0.67) and pH (-0.68); free with total sulfur dioxide (+0.67); citric acid with volatile acidity (-0.55); and alcohol with density (-0.50, because alcohol is lighter than water). These links warn about multicollinearity in the regression.

3.3 Attributes most associated with quality (Task 2d)

Attribute	r with quality
alcohol	+0.476
volatile acidity	-0.391
sulphates	+0.251
citric acid	+0.226
total sulfur dioxide	-0.185
density	-0.175
chlorides	-0.129
fixed acidity	+0.124
pH	-0.058
free sulfur dioxide	-0.051
residual sugar	+0.014

Ranked by absolute Pearson correlation, the two strongest are **alcohol (+0.476, positive)** and **volatile acidity (-0.391, negative)**. Higher alcohol goes with higher quality. Volatile acidity is mainly acetic acid, which gives a vinegar smell and taste, so tasters mark those wines down. Sulphates (+0.25) and citric acid (+0.23) are weaker, and residual sugar (0.01), free sulfur dioxide (-0.05) and pH (-0.06) are almost unrelated to quality on their own.

4. Model Evaluation

Amultiple linearregression using all 11 attributes was fitted on 80% of the data (1279 wines) and tested on the remaining 20% (320 wines), random_state = 42.

Metric	Test value
MAE	0.5035
MSE	0.3900
RMSE R-	0.6245
squared	0.4032

Explanation. The MAE of 0.50 means predictions are off by about half a quality point on average. The RMSE of 0.62 is a little larger because it penalises big errors more. R^2 = 0.40 means the model explains only about 40% of the variation in quality of unseen wines; 60% is unexplained. Always predicting the training mean would give an RMSE of 0.81, so the model is better than guessing but not by a large margin. Train R^2 (0.35) is slightly below test R^2, so there is no sign of overfitting; the test split was probably a little easier, so a single split should not be over-trusted. On a scale where scores are mostly 5 or 6, an error of half a point is substantial.

4.1 Most influential predictors and multicollinearity (Task 3d)

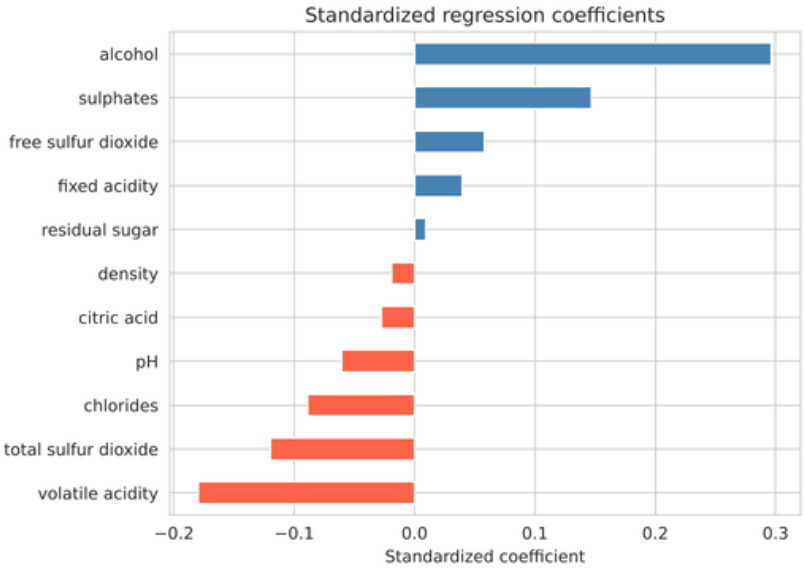


Figure 4: Standardized regression coefficients (x-axis: change in quality per 1 std increase in the attribute).

Interpretation of Figure 4. Raw coefficients cannot be compared because the attributes have different units, so the features were standardized. The two most influential predictors are **alcohol (+0.297)** and **volatile acidity (-0.179)**; sulphates (+0.147) is third. This matches the simple correlations in Section 3.3.

Attribute	VIF
fixed acidity	7.37
density	6.02
pH	3.28
citric acid	3.21
alcohol	2.93
total sulfur dioxide	2.20
free sulfur dioxide	1.98
volatile acidity	1.81
residual sugar	1.73
chlorides	1.52
sulphates	1.46

Multicollinearity. A VIF above 5 deserves attention and above 10 is serious. Fixed acidity (7.37) and density (6.02) exceed 5, which fits the heatmap, but no VIF is above 10, so the problem is moderate. Alcohol (2.93) and volatile acidity (1.81) have low VIFs, so their coefficients can be interpreted with reasonable confidence: higher alcohol is associated with higher quality, and higher volatile acidity with lower quality, holding the other attributes fixed. The coefficients of fixed acidity and density should not be read literally because they are entangled with each other and with alcohol.

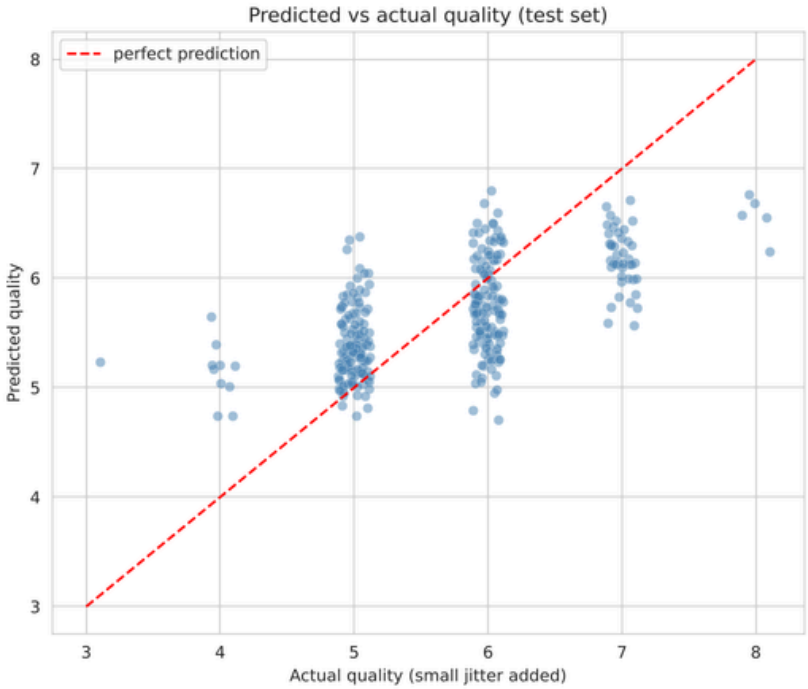


Figure 5: Predicted vs actual quality on the test set (jitter added to actual values; dashed line = perfect prediction).

Interpretation of Figure 5 and limitations. If the model were perfect all points would lie on the dashed line. Instead the cloud is flatter: predictions only range from 4.7 to 6.8, so the model never predicts a 3 or an 8. Poor wines are over-predicted (actual 3s and 4s average about 5.1 to 5.2) and the best wines are under-predicted (actual 7s average about 6.2 and actual 8s about 6.6). The scatter is wide at every level; only 57% of test wines are predicted within 0.5 of the true score and 89% within 1 point. Linear regression also assumes constant straight-line effects, cannot capture interactions or thresholds, and treats an ordered whole-number score as a continuous value.

5. Business Decision-Making and Conclusion

5.1 Measurements to focus on

Alcohol is the strongest correlate of quality (r = +0.48), has the largest standardized coefficient (+0.30), a low VIF, and is cheap and accurate to measure. In this data about two thirds of wines below 10% alcohol are rated 5, while among wines at 11.5% or more only about 10% are rated 5 or lower, so a low reading is a useful flag. **Volatile acidity** is second (r = -0.39, standardized coefficient -0.18), has a low VIF and a clear meaning (acetic acid, vinegar taint), so a high reading is a sensible reason to hold a batch for tasting. Sulphates would be the next measurement to add.

5.2 Can the model replace sensory evaluation?

No. R^2= 0.40 leaves 60% of the variation unexplained. RMSE = 0.62 and MAE = 0.50 are large when commercial grades differ by one or two points. Only 57% of predictions are within 0.5 of the true score and about 11% are off by more than a full point. The model also cannot identify the best and worst wines, which are the ones the company cares about most. It is better used as a cheap first screening step to flag unusual batches, while the sensory panel makes the final decision.

5.3 Limitations

1. Subjective, imbalanced and duplicated data. Each score is the median of at least three experts, so it contains human noise and taste that no chemical measurement can predict. About 83% of wines are rated 5 or 6 with very few 3s and 8s. The 240 duplicate rows may also make the test score slightly optimistic because the same row can appear in both train and test sets.

2. Limited scope and a simple model. The data covers only red Vinho Verde from one region, with no information on grape, vineyard, vintage, brand or price, so results should not be generalised to other wines. Only linear regression and a single train-test split were used, so the numbers could change somewhat with another split or with non-linear models.

5.4 Conclusion

Alcohol (positive) and volatile acidity (negative) are the two attributes most associated with red wine quality. A linear regression on all 11 measurements reaches R^2 of about 0.40 and RMSE of about 0.62 on unseen wines. That is useful for screening but not accurate enough to replace the sensory panel. Future work could try random forest or gradient boosting, treat the task as classification (good vs not good), use cross-validation, and compare with the white wine data.

References and acknowledgement

P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis. Modeling wine preferences by data mining from physicochemical properties. Decision Support Systems, Elsevier, 47(4):547-553, 2009.